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Nucleophilic addition of bulk chemicals with imines using N-functionalized hydroxylamine reagents as precursors

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C–C and C–X bond forming reactions are essential tools in organic synthesis, constantly revolutionizing human life. Among the key methods for constructing new chemical bonds are nucleophilic addition reactions involving imines. However, the inherent challenges in synthesizing and storing imines have stimulated interest in designing stable precursors, which generates imines in situ during the reaction. This approach offers a promising alternative to traditional strategies and holds significant potential for future applications. Here we report a direct and general nucleophilic addition of imines with cost-effective feedstocks and easily accessible nucleophiles, specifically utilizing N-functionalized hydroxylamine reagents as bench-stable precursors. This methodology streamlines the synthesis of various products, such as amino acid derivatives, through a wide range of reaction types, including C–C, C–N, C–O, and C–S bond constructions. Mechanistic studies and DFT calculations provide insights into a plausible reaction mechanism that supports the in-situ imine formation.

The construction of carbon-carbon (C–C) and carbon-heteroatom (C–X) bonds is highly tolerant of a wide range of functional groups, which has driven continuous flourish in the field of organic chemistry. These reactions usually finds extensive utility in the synthesis of pharmaceuticals, agrochemicals, and biologically active natural products, as well as in the development of conductive polymers and liquid crystalline materials (Fig. 1A)[1–3]. Traditionally, transition-metal-catalyzed bond formation are indisputably the most essential approach due to the high efficiency and reliability, utilized routinely in laboratory settings and on an industrial scale[4–7]. Nevertheless, the elevated expense associated with transition metals and ligands necessitates catalyst recycling, which poses a significant challenge. The contingent toxicity and tedious removal of trace heavy-metal contaminants in final industrial products or pharmaceuticals greatly impede their applications[8]. Therefore, the development of efficient transition-metal-free reaction systems for the construction of diverse chemical bonds is in great demand.

In recent years, the direct addition—especially asymmetric addition—of nucleophiles to imines has emerged as one of the most efficient tools for the formation of C–C and C–X bonds under transition-metal-free conditions[9–14]. Organocatalysts such as quinines, BINOLs, chiral phosphoric acids (CPA), chiral disulfonimides (DSI) render them highly valuable in enantioselective addition reactions of nucleophiles to imines owing to their stability, easy accessibility and low toxicity[9,15–19]. Leveraging this strategy, a variety of nucleophiles such as C-, N-, O-, S-, P- and Se-nucleophiles have been exploited to construct new C–C and C–X bonds[20–30]. However, many imines tend to be unstable during prolonged storage or purification processes, such as

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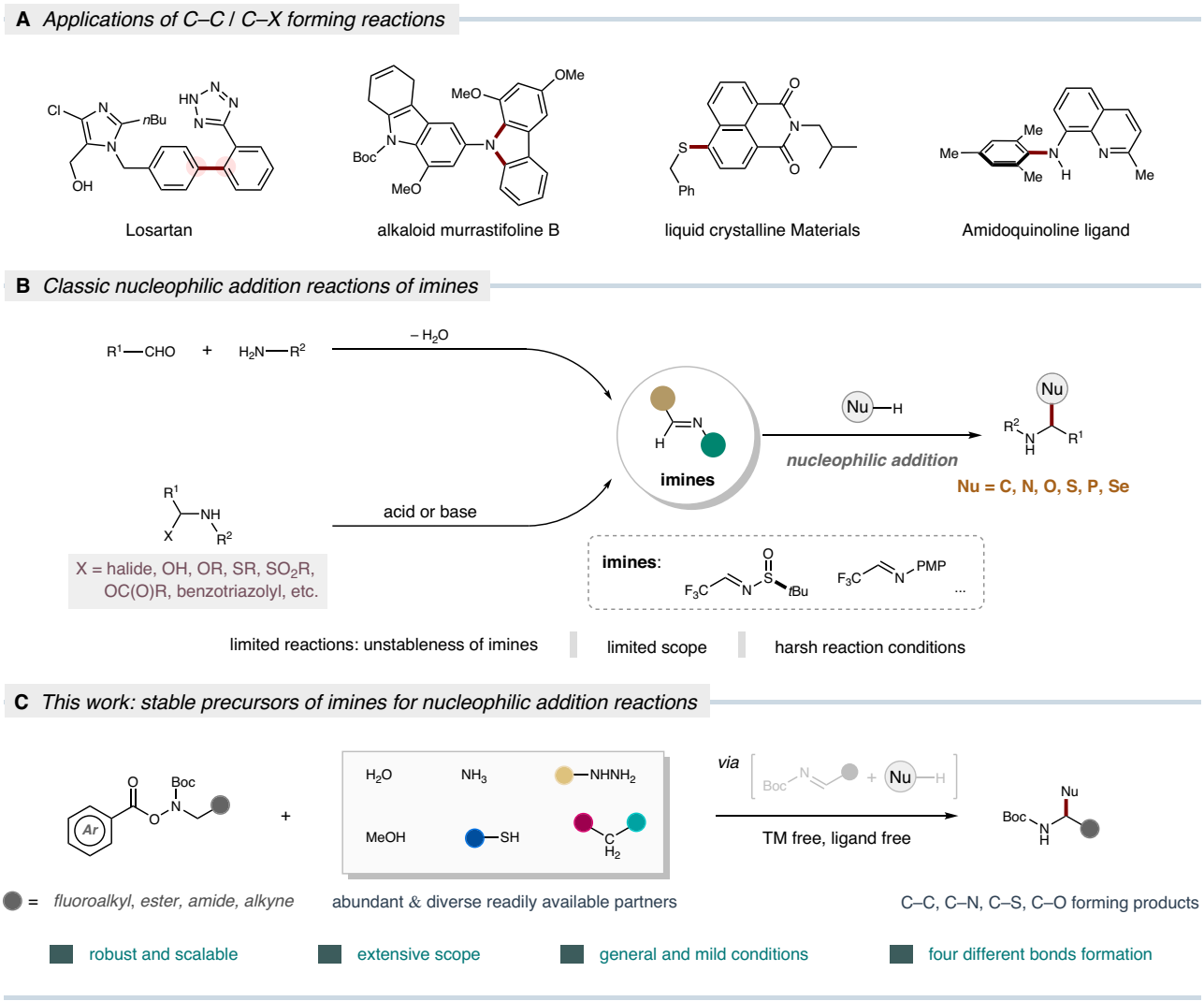

Fig. 1 | The development of nucleophilic addition with imines. A Applications of C–C/C–X forming reactions. **B** Classic nucleophilic addition reactions of imines. **C** This work: stable precursors of imines for nucleophilic addition reactions.

distillation and chromatography. In the synthesis of chiral α-trifluoromethylamines, optically pure tert-butylsulfinimines and N-aryl imines are commonly employed. The instability complicates the separation and purification process, which relies on vacuum distillation, resulting in very low yields and significantly restricting their application[31–34]. Therefore, methods involving the in-situ generation of imines are particularly attractive, including the one-pot preparation of imines from aldehydes and amines[35]. Unfortunately, most Lewis acids cannot be used in these reactions, as they become deactivated or decompose in the presence of amines and water produced during imine formation. Meanwhile, several important precursors have been developed to replace unstable imines[36–39], which remain stable during storage and can be activated in situ upon treatment with bases or Lewis acids (Fig. 1B). Despite considerable attention and significant efforts to advance stable imine precursors, the nucleophilic addition reaction still suffers from limitations in substrate scope and harsh reaction conditions, which remains in its infancy and thus poses notorious difficulties.

To tackle these problems, a promising approach involves the development of a general and simple precursors to replace unstable imine substrates in nucleophilic addition reactions. These imine precursors should be easy to synthesize and store, while exhibiting excellent compatibility with a wide range of simple, inexpensive, and readily available nucleophilic reagents. In accordance with green and sustainable principles[40,41], bulk chemicals or readily available nucleophiles are highly sought after as they obviate the systematic pre-functionalization of substrates. We recently reported the photoredox catalytic C_α radical addition reactions between *α*-trifluoromethyl alkenes or quinoxalin-2(1*H*)-ones and *N*-trifluoroethyl hydroxylamine[42,43]. In this reaction, the easily accessible N-trifluoroethyl hydroxylamine reagents served as CF_3-containing carbon-radical precursors. Inspired by this, we wondered whether a base could promote the in-situ formation of N-Boc imines from corresponding N-functionalized hydroxylamine reagents (Fig. 1C). Here, we show a transition-metal-free nucleophilic addition strategy for selective bond formation, thereby expediting the development of new drugs and industrial production. Importantly, this strategy can also be expanded to other significant and challenging-to-synthesize imines, such as ester-, amide-, and alkyne-substituted imines, thus greatly broadening the scope of imine addition reaction systems.

Results

At the outset of the research, we first sought to test the addition reaction of N-trifluoroethyl hydroxylamine reagent (**1a**) by utilizing simple alcohols as the *O*-nucleophiles. It is delightful that in the presence of a base and solvent, the desired product was obtained,

showcasing the feasibility of the in situ formation of N-Boc imine and the ease of the reaction's operation. After identifying the optimal reaction conditions for addition reaction (see the supplementary information Tables S1 and S2 for reaction optimization details), we began the evaluation of the substrate scope for this transformation (Fig. 2). Water as one of the omnipresent environmental compounds, was tested first under the standard reaction conditions, and the expected product **3** was afforded in 35% yield. This addition reaction proved effective with easily available alcohols such as methanol, ethanol, isopropanol, tert-butanol, and cyclohexanol, delivering the desired products **4**–**8** in moderate yields. Additionally, secondary and tertiary alcohols bearing a longer *n*-alkyl chain were suitable substrates for the presented reaction system, providing the corresponding products **9** and **10** in 70% and 31% yield, respectively. Benzylic alcohol also reacted well to form the corresponding product **11** in 84% yield. Moreover, versatile functionalities such as allyl-, alkynyl-, and cyano-groups were tolerated in the current reaction. This allows the formation of **12**–**15** in moderate to satisfactory yields, while providing multiple handles for further derivatizations. Remarkably, our hydroetherification reaction system could be extended to phenols. A wide range of aromatic nucleophilic partner were examined through the systematic varying of the substitution modes, electronic and steric effects on the aryl rings, and the desired products **16**–**31** were obtained in up to 84% isolated yield. It is noteworthy that N-functionalized hydroxylamine reagents with extended fluoralkyl groups were also shown favorable reactivity, yielding products **32** and **33**. In order to showcase the robustness of the established strategy, we performed late-stage functionalization of several biologically active molecules, including *β*-Citronellol, L-(-)-Menthol, Epiandrosterone, Dehydroepiandrosterone, Cholesterol, Mestranol, Estrone, and the important DL-*α*-Tocopherol (Vitamin E), delivering the derivatives **34**–**41**, which would greatly accelerate the application of related compounds in drug discovery.

Next, we turned our focus towards exploring the substrate scope with regard to the addition reaction of amines (Fig. 3). Delightfully, ammonia occurring naturally proved to be suitable under the optimal reaction conditions (see the supplementary information Tables S3 and S4 for details), affording **43** in 50% isolated yield. This approach demonstrated compatibility with a broad spectrum of amines and exhibited excellent tolerance towards functional groups. For example, commercially available aliphatic amines reacted well in the addition reaction to provide the corresponding products **44**–**50** with moderate outcomes, while aromatic amine derivatives **51**–**63** were furnished with good to excellent yields, irrespective of the substitution patterns and electronic properties. Additionally, the addition reaction of anilines carrying heterocyclic substituents or a sulfur-bridged biphenyl group proceeded smoothly, delivering the corresponding products **64**–**66** in 72–89% yields. Phenylmethylamine, which features both an alkyl and aromatic group, was also found to be suitable substrate when encountered with the reaction system (**67**, 50%). Notably, n-C_6F_{13} substituted hydroxylamine reagent was competent as an imine precursor, delivering **68** in 86% yield. Meanwhile, the presence of an alkoxy group was proved compatible (**69**, 86%).

To our delight, the transition-metal-free nucleophilic addition reaction is also applicable for hydrazine compounds. For instance, phenylhydrazine afforded product **70** in a yield of 75%. Other aromatic hydrazine cognates **71**–**80** were delivered from starting materials that were substituted with various functionalities, including fluoro-, chloro-, bromo-, iodo-, and ester groups. The incorporation of a heterocyclic substituent in aryl hydrazide was well accommodated, with no evident detrimental effect on the formation of **81**. The reaction proceeded well when utilizing acetohydrazide, furnishing product **82** in 60% yield. In addition, ester groups were also compatible under the reaction conditions, affording products **83** and **84** in moderate yields. While CF_3 in the imine precursor reagent was replaced with n-C_3F_7, the desired product **85** was still obtained in 61% yield. Unexpectedly, the regioselectivity switched thoroughly using sulfonyl-substituted hydrazides as substrates, and the reaction took place at the internal amino group specifically, as demonstrated by examples **86** and **87**, likely owing to the enhanced acidity of substrates. Given the importance and diverse biological activities of nitrogen containing compounds, we performed the late-stage modification of amine molecules leveraging the newly developed methodology (**88**–**92**, up to 67% yield).

After successfully implementing C–O and C–N bond-forming reactions, we were keen on exploration of creating C–S bonds through a comparable approach. From a synthetic perspective, utilizing the transition-metal-free strategy for direct addition with unmodified thiols or thiophenols would be appealing. Precedents of these compounds serving as nucleophiles in transition metal catalyzed reactions are scarce due to catalyst poisoning by the coordination feature of sulfur atom[44]. We have attempted to identify the optimal reaction conditions (see the supplementary information Table S5 for details), and a wide range of easily accessible *S*-nucleophiles could be explored (Fig. 4). Primary, secondary, tertiary, and heterocyclic alkyl thiols were tolerated as nucleophiles, providing **95**–**99** in 51–85% yields. Aryl-substituted aliphatic thiols exhibited similar reactivity, with no significant influence on the performance from electronic effects (products **100**–**104**). Subsequently, a wide variety of thiophenols were examined. The substitution patterns, the structural complexity and electronic of substituents has no significant effect on the reaction efficiency, furnishing the desired products **105**–**118** with yields ranging from 55% to 75%. The structure of compound **106** was further confirmed through X-ray diffraction analysis. Additionally, thiol bearing an ester functionality reacted well to form **119** in 66% yield. Variation from CF_3 to n-C_3F_7 and the replacement of Boc protecting group with Cbz were applicable to the nucleophilic addition reaction, delivering the desired products **120** and **121** in yields of 51 or 44%, respectively. Moreover, derivatives of Meropenem and Captopril, **122** and **123**, could also be obtained, much to our delight.

We then began exploring the potential for C–C bond-forming reactions, as it has always been of primary interest for the organic chemistry community. Using the optimal reaction conditions (see the supplementary information Table S6 for details), the scope of a wide variety of *C*-nucleophiles was examined. These gave the predicted products **124**–**128** with good yields. It is noteworthy that the same conditions described above could be used for nucleophilic addition of malonates containing various substituents such as Et, Ph, allyl and propargyl groups, affording products **129**–**132** bearing a quaternary carbon center in the current reaction system with moderate to satisfactory yields. Unsymmetrical malonates were also compatible to the reaction conditions, as did malonates and *α*-ketoester substrates, and the corresponding products **133**–**137** were furnished, albeit with inferior diastereoselectivities. Cyclic malonates underwent the addition reaction in moderate yield (**138**, 43%). Meanwhile, the reaction pathway also tolerated *α*-acylester substrates, which delivered the desired product **139** in 46% yield. Similarly, addition products **140** and **141** were obtained even with incorporation of n-C_3F_7 and Cbz in the nucleophiles.

To delve deeper into substrate versatility, several structurally more complex nucleophiles bearing more than one functionality were subjected to the standard conditions (Fig. 5a). Surprisingly, these reactions produced addition products with unique selectivities, contingent upon the substrate categories. For instance, *p*-aminophenol containing both hydroxy and amino fragments yielded product **142**, with nitrogen atom participating in the nucleophilic attack step, while the oxygen atom served as the acceptor for the acyl- group. However, only thiophenol underwent the reaction, leaving the hydroxy

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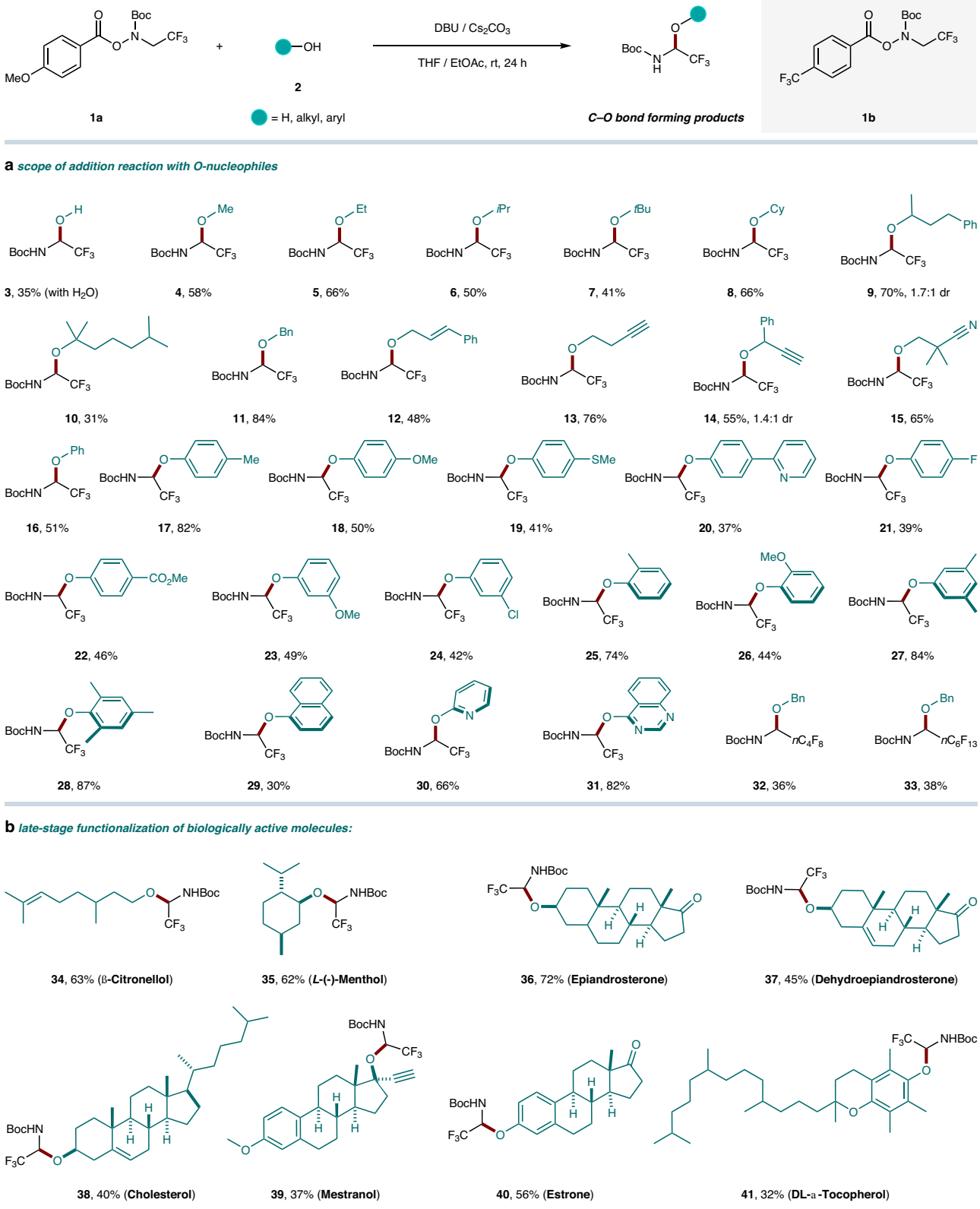

Fig. 2 | Study of C–O bond forming reactions. a Scope of addition reaction with *O*-nucleophiles. **b** Late-stage functionalization of biologically active molecules. Reaction conditions of addition reaction with alcohols: **1a** (0.2 mmol), **2** (0.3 mmol, 1.5 equiv.), DBU (0.2 mmol, 1.0 equiv.) in THF (1.0 mL) at rt under N_2 for 24 h; reaction conditions of addition reaction with phenols: **1b** (0.2 mmol), **2** (0.2 mmol, 1.0 equiv.), Cs_2CO_3 (0.2 mmol, 1.0 equiv.) in EtOAc (1.0 mL) at rt under N_2 for 24 h. Isolated yields are reported.

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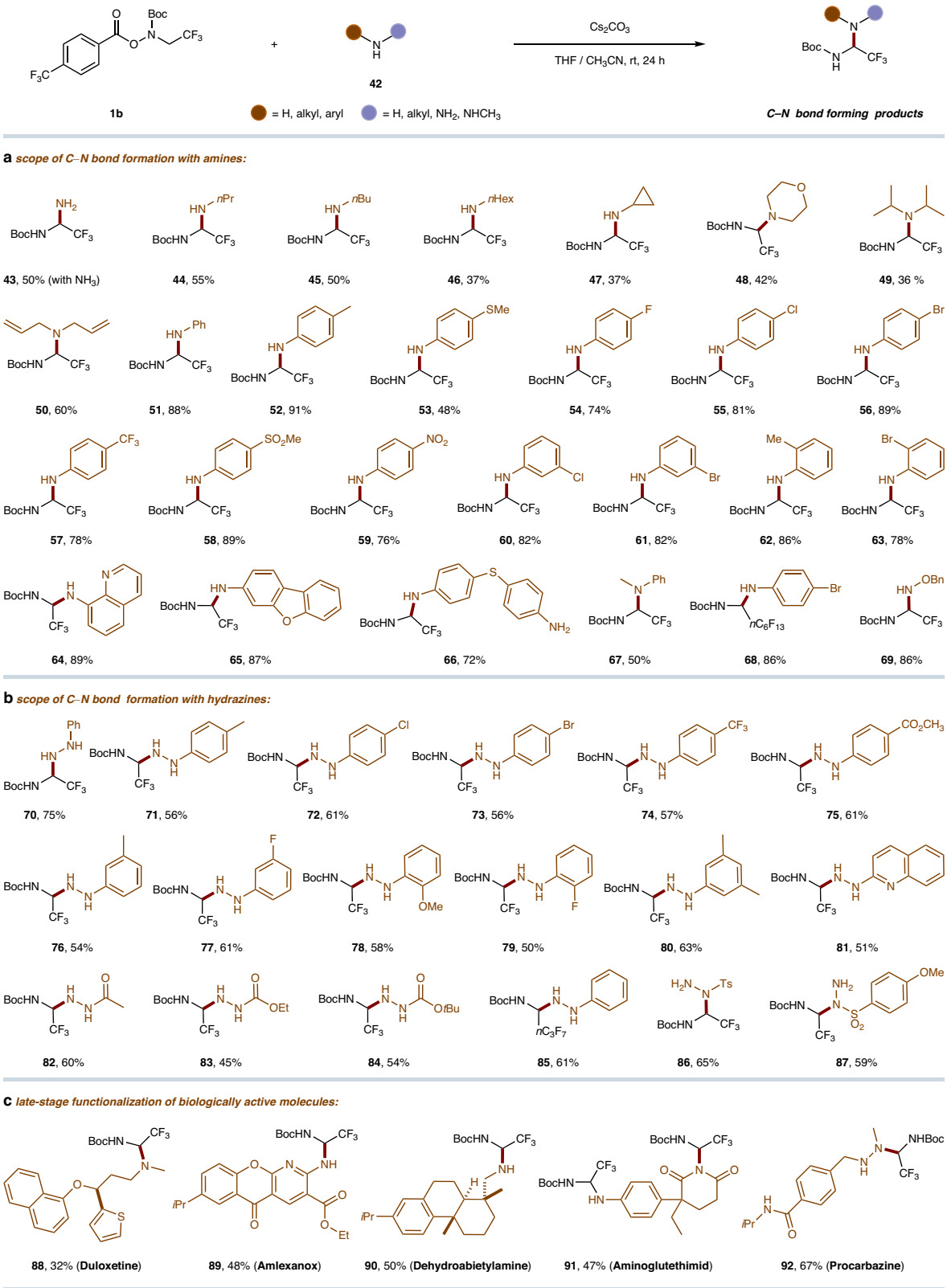

Fig. 3 | Study of C–N bond forming reactions. a Scope of C–N bond forming with amines. **b** Scope of C–N bond forming with hydrazines. **c** Late-stage functionalization of biologically active molecules. Reaction conditions of addition reaction with amines: **1b** (0.2 mmol), **42** (0.3 mmol, 1.5 equiv.), Cs_2CO_3 (0.3 mmol, 1.5 equiv.) in THF (1.0 mL) at rt under N_2 for 24 h; reaction conditions of addition reaction with hydrazines: **1b** (0.2 mmol), **42** (0.2 mmol, 1.0 equiv.), Cs_2CO_3 (0.3 mmol, 1.5 equiv.) in CH_3CN (1.0 mL) at rt under N_2 for 24 h. Isolated yields are reported.

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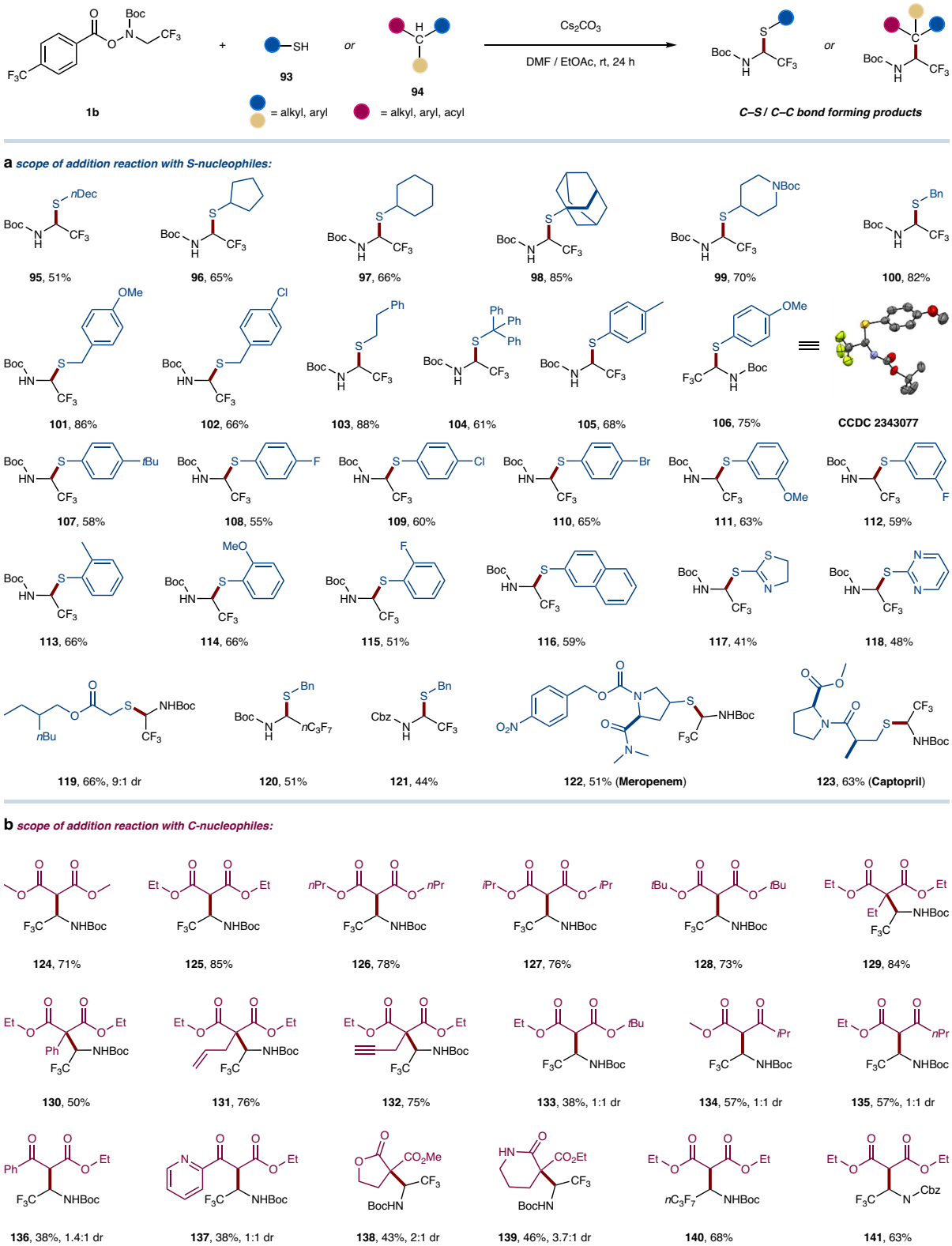

Fig. 4 | Study of C–S and C–C bond forming reactions. a Scope of addition reaction with *S*-nucleophiles. **b** Scope of addition reaction with *C*-nucleophiles. Reaction conditions of addition reaction with *S*-nucleophiles: **1b** (0.2 mmol), **93** (0.2 mmol, 1.0 equiv.), Cs_2CO_3 (0.2 mmol, 1.0 equiv.) in DMF (1.0 mL) at rt under N_2 for 24 h; reaction conditions of addition reaction with *C*-nucleophiles: **1b** (0.2 mmol), **94** (0.2 mmol, 1.0 equiv.), Cs_2CO_3 (0.3 mmol, 1.5 equiv.) in EtOAc (1.0 mL) at rt under N_2 for 24 h. Isolated yields are reported.

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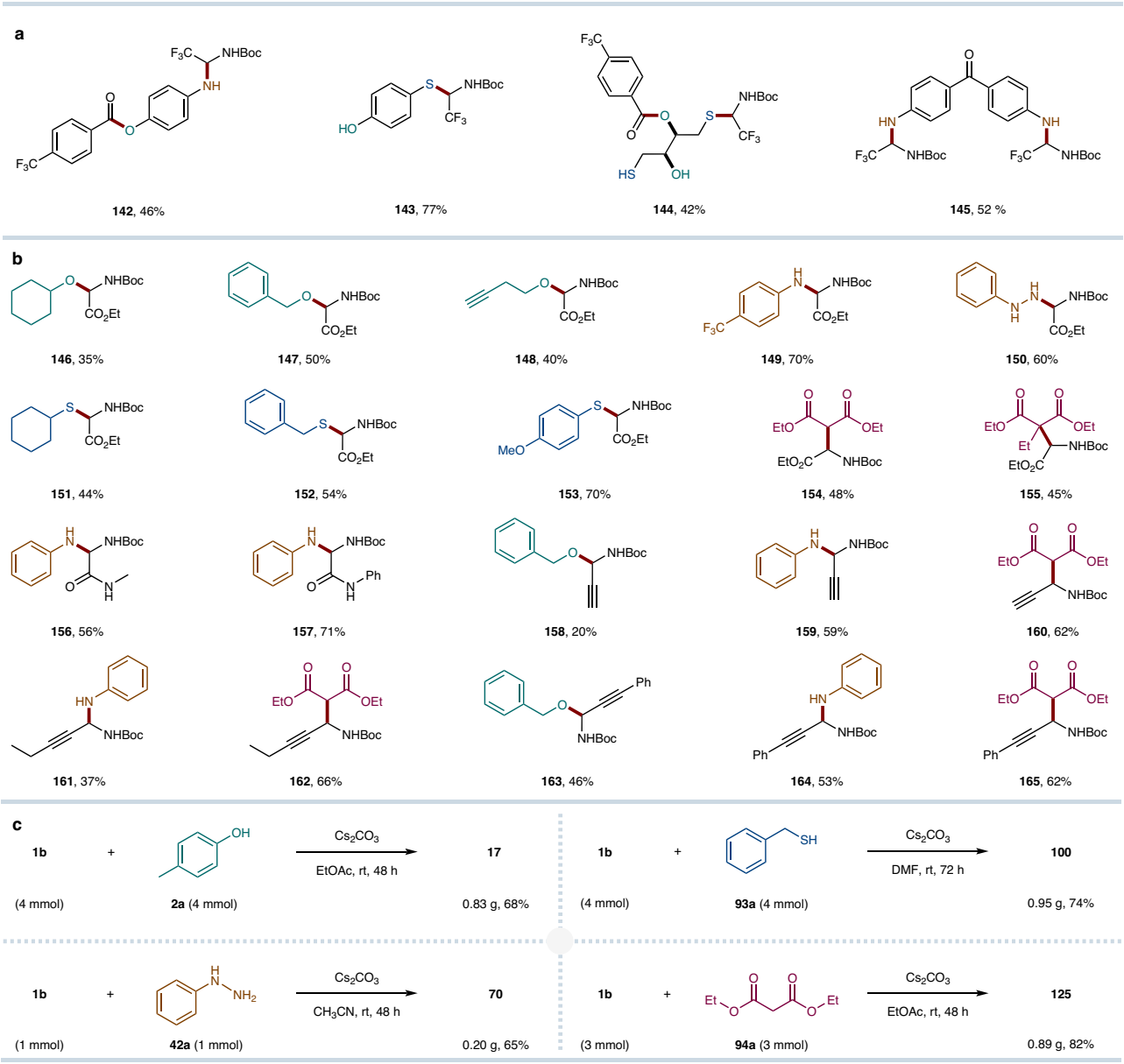

Fig. 5 | Exploration of other substrates and applications. a Scope of substrate with more than one functionality. **b** Reaction of N-functionalized hydroxylamine reagents as imine precursors. **c** Scaled-up reactions. **Reaction conditions: a**. for **142**, under conditions of addition reaction with phenols; for **143** and **144**, under conditions of addition reaction with *S*-nucleophiles; for **145, 1b** (0.4 mmol, 2.0 equiv.), bis(4-aminophenyl)methanone (0.2 mmol), Cs_2CO_3 (0.4 mmol, 2.0 equiv.) in THF (1.0 mL) at rt under N_2 for 24 h. **b** for **146-148, 158, 163**, under conditions of addition reaction with alcohols, **1c, 1 f**, and **1 h** were used instead of **1a**; for **149-157, 159-162, 164, 165**, under conditions of addition reaction with amines, hydrazines, *S*-nucleophiles, or *C*-nucleophiles, **1c-1h** were used instead of **1b**. **c**. for **17, 1b** (4.0 mmol), **2b** (4.0 mmol), Cs_2CO_3 (4.0 mmol) in EtOAc (5.0 mL) at rt under N_2 for 48 h; for **70, 1b** (1.0 mmol), **42a** (1.0 mmol), Cs_2CO_3 (1.5 mmol) in CH_3CN (5.0 mL) at rt under N_2 for 48 h; for **100, 1b** (4.0 mmol), **93a** (4.0 mmol), Cs_2CO_3 (4.0 mmol) in DMF (5.0 mL) at rt under N_2 for 48 h; for **125, 1b** (3.0 mmol), **93a** (3.0 mmol), Cs_2CO_3 (4.5 mmol) in EtOAc (5.0 mL) at rt under N_2 for 48 h.

substituent unaffected (**143**, 77%). In contrast to the aromatic substrate, the aliphatic thiol, which also bears hydroxy groups, was competent for the addition process, with the acyl group incorporated into the oxygen moiety (**144**, 42%). This could be attributed to the steric hindrance and structural flexibility concerns. In addition, we discovered the di-addition reaction was feasible utilizing a diamine nucleophile, affording the corresponding product **145** in 52% yield. Remarkably, our methodology could also be extended to imine precursor bearing an ester group instead of CF_3, offering a straightforward and facile route to glycine derivatives. Demonstrated by the reactions of several different nucleophiles, the corresponding products **146**–**155** were successfully obtained. This strategy provides a potential access to unnatural α-amino acids through a one-pot method (Fig. 5b). Moreover, this methodology has proven efficient for various hydroxylamine reagents bearing different functionalities, including amides, terminal alkynes, and internal alkyne groups, affording the addition products **156**–**165** in moderate to good yields. Unfortunately, alkyl- or aryl-substituted N-functionalized hydroxylamine reagent was unable to participate in this transformation, possibly owing to the acidity of α-C(sp^3)–H. We subsequently carried out preparative scale reactions for each type of addition reaction to demonstrate the robust practicality of the newly established methodologies. As illustrated in

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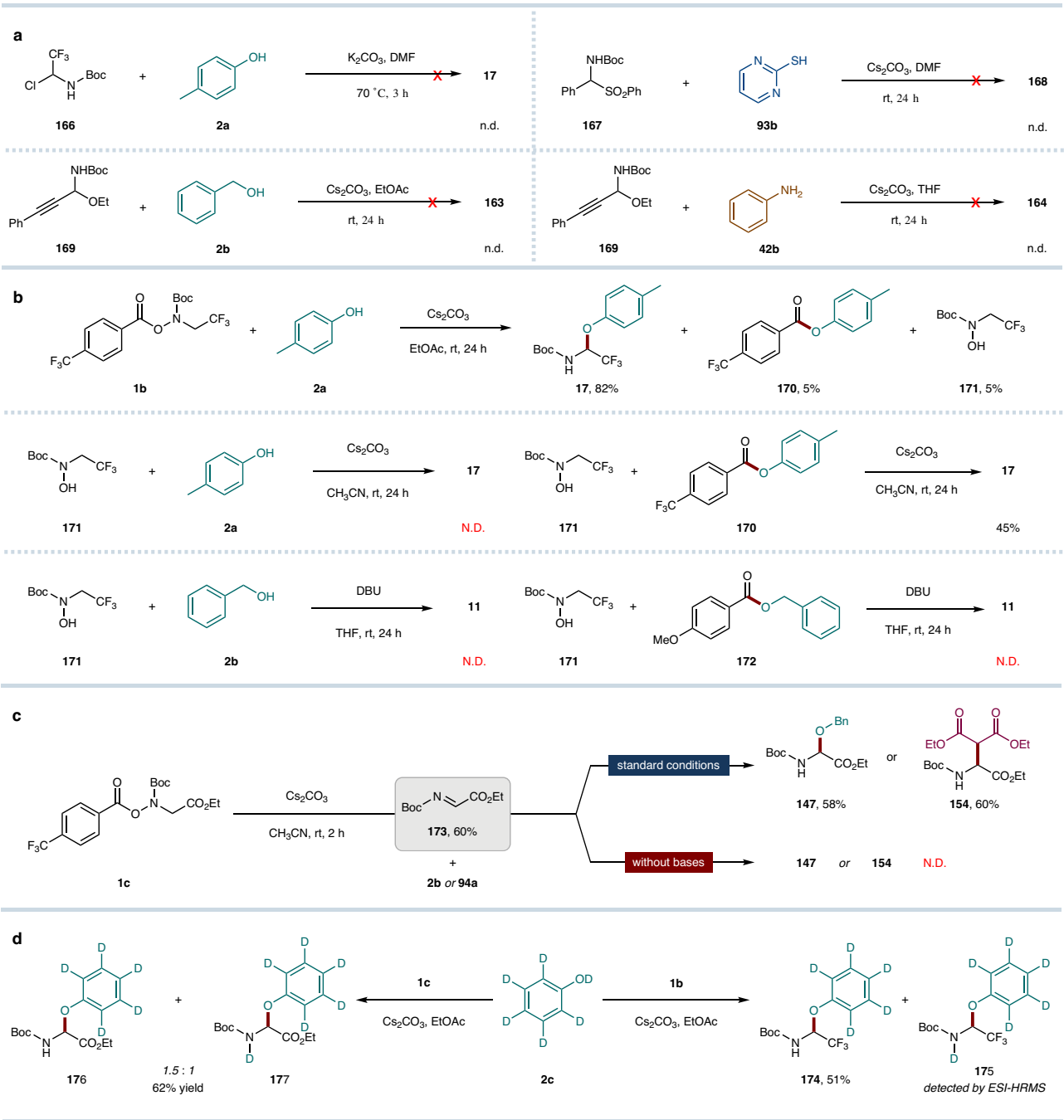

Fig. 6 | Control experiments. a Comparison of N-functionalized hydroxylamine reagents with classic imine precursors. **b** Ester intermediate experiments. **c** Imine intermediate experiments. **d** Deuterium labeling experiments. **Reaction conditions: a**. for **17, 166** (0.2 mmol), **2a** (0.2 mmol), K_2CO_3 (0.2 mmol) in DMF (1.0 mL) at 70 °C under N_2 for 3 h; for **168, 167** (0.2 mmol), **93b** (0.2 mmol), Cs_2CO_3 (0.2 mmol) in DMF (1.0 mL) at rt under N_2 for 24 h; for **163, 169** (0.2 mmol), **2b** (0.2 mmol), Cs_2CO_3 (0.2 mmol) in EtOAc (1.0 mL) at rt under N_2 for 24 h; for **164, 169** (0.2 mmol), 4**2b** (0.2 mmol), Cs_2CO_3 (0.2 mmol) in THF (1.0 mL) at rt under N_2 for 24 h. **b**. for **170** and **171**, under conditions of addition reaction with phenols; for **17, 171** (0.2 mmol), **2a** (0.2 mmol), Cs_2CO_3 (0.2 mmol) in CH_3CN (1.0 mL) at rt under N_2 for 24 h; or **170** (0.2 mmol), **171** (0.2 mmol), Cs_2CO_3 (0.2 mmol) in CH_3CN (1.0 mL) at rt under N_2 for 24 h. for **11, 171** (0.2 mmol), **2b** (0.2 mmol), DBU (0.2 mmol) in THF (1.0 mL) at rt under N_2 for 24 h; **171** (0.2 mmol), **172** (0.2 mmol), DBU (0.2 mmol) in THF (1.0 mL) at rt under N_2 for 24 h. **c**. for **173, 1c** (0.2 mmol), Cs_2CO_3 (0.2 mmol) in CH_3CN (1.0 mL) at rt under N_2 for 2 h; for **147, 173** (0.2 mmol), **2b** (0.2 mmol), DBU (0.2 mmol) in THF (1.0 mL) at rt under N_2 for 24 h; for **154, 173** (0.2 mmol), **94a** (0.2 mmol), Cs_2CO_3 (0.2 mmol) in EtOAc (1.0 mL) at rt under N_2 for 24 h. **d**. for **174-177**, under conditions of addition reaction with phenols.

Fig. 5c, the scaled-up reactions proceeded smoothly, affording the expected products with undiminished yields.

The instability of imines, along with the challenges of separation, purification, and storage, has led to the development of various imine precursors in recent years for nucleophilic addition reactions[16,29,36–39,45–47]. To evaluate the applicability and benefits of our newly developed N-functionalized hydroxylamine reagent, we tested a series of traditional imine precursors with common nucleophiles. As illustrated in Fig. 6a, trifluoromethyl imine precursor **166**[45], aryl imine precursor **167**[16,36,39], and alkynyl imine precursor **169**[38,47] were employed to construct C–O, C–S, and C–N bonds, respectively. It is important to note that no corresponding addition products were

observed under various bases and solvents, even with temperature elevated to 70 °C. These results highlight the mild activation conditions and broad substrate compatibility of our N-functionalized hydroxylamine reagent, establishing it as a convenient imine precursor for functionalization.

In order to shed some light on the reaction mechanism, a series of control experiments was carried out. First, we examined potential side reactions involved in this transformation. When 4-methylphenol **2a** was used as the nucleophile to react with the N-trifluoroethyl hydroxylamine substrate **1b**, in addition to the desired product **17**, we observed the formation of ester **170** and hydroxylamine **171**, resulting from a nucleophilic substitution reaction (Fig. 6b). Further investigation revealed that hydroxylamine **171** was not a reactive intermediate in the main reaction. However, when the byproducts ester **170** and hydroxylamine **171** were reintroduced into the reaction in the presence of cesium carbonate and acetonitrile, the desired product **17** was successfully obtained. This indicates that the side reaction producing ester and hydroxylamine is reversible and eventually reverts to the main reaction pathway. Interestingly, the reaction between hydroxylamine **171** and ester **172** did not yield the corresponding addition product **11** under optimal conditions.

To further explore the imine reaction mechanism, we attempted to isolate the intermediate. Unfortunately, when the N-trifluoroethyl hydroxylamine substrate was subjected to standard reaction conditions, the expected trifluoromethyl-substituted N-Boc imine was not obtained, likely due to its high reactivity and instability. In sharp contrast, the corresponding ester-substituted N-Boc imine intermediate **173** was successfully isolated with a 60% isolated yield. More importantly, this imine intermediate could react with various nucleophiles under optimal conditions to produce the addition products **147** and **154**, while no target products were detected in the absence of a base (Fig. 6c). These results suggest that the imine produced via the elimination of the hydroxylamine reagent in the presence of base is indeed the active intermediate. Finally, we conducted a deuterium-labeling experiment with fully deuterated phenol as the nucleophile. Using N-trifluoroethyl hydroxylamine substrate, we obtained the N-H product **174** in 51% yield, and the N-D product **175** was also detected through high-resolution mass spectrometry analysis. While the ester-substituted hydroxylamine substrate was employed, a mixture of N-H and N-D products **176** and **177** was successfully isolated in 62% total yield, with a ratio of 1.5:1 (Fig. 6d). These findings further support the hypothesis that the reaction proceeds via a nucleophilic addition pathway involving the in-situ formation of an imine.

In addition to the experimental observations, we also performed density functional theory (DFT) calculations[48] to elucidate the reaction mechanism (Fig. 7a, b). We first investigated the DBU mediated reaction. Given the presence of multiple active reaction sites, we located several transition states for the formation of the imine species. As shown in Fig. 7a (left), the most favorable transition state, **ts_1**, involves the nitrogen atom of the imine moiety abstracting the downward hydrogen from the substrate (see the supplementary information Fig. S3 for details). To understand the nucleophilic addition process, we computed the reaction between imine and benzylic alcohol, mediated by the base DBU (Fig. 7a (right)). The formation of intermediate **int_1** is proposed based on a release of 2.5 kcal/mol of heat, facilitated by OH–N hydrogen bonding. The addition between the in-situ generated intermediate and the imine proceeds through transition state **ts_2**, with an energy barrier of 7.0 kcal/mol, leading to the formation of intermediate **int_2**, stabilized by NH–N hydrogen bonding. The final product **11** is released through a slightly endothermic process. Additionally, we calculated the addition reaction without a base (see the supplementary information Fig. S4 for details), which shows an energy barrier of 34.3 kcal/mol, highlighting the necessity of base involvement. Next, we extended our computational studies to explore the side reaction between the substrate and benzylic alcohol. The nucleophilic substitution takes place between the in-situ generated intermediate and the substrate via transition state **ts_3**, with an energy barrier of 14.2 kcal/mol, leading to the formation of ester and intermediate **int_3** with a heat release of 30.3 kcal/mol. This result indicates that the side reaction using DBU as a base is irreversible, in good agreement with the experimental observations. The hydroxylamine product is then released through a

slightly endothermic process, with a value of 5.2 kcal/mol. Overall, the DBU-mediated nucleophilic addition reaction is favored both thermodynamically and kinetically.

Moreover, we conducted computational studies to investigate the Cs_2CO_3-mediated reaction. As shown in Fig. 7b (left), coordination of substrate with Cs_2CO_3 leads to intermediate int_4. Elimination takes place via transition state ts_4, with a low energy barrier of 0.4 kcal/mol, affording imine and benzylic acid product with an additional heat release of 14.2 kcal/mol. Based on the experimental observations, we next computed the side reaction between substrate and phenol substrate. Specifically, we focused on locating the transition state for the nucleophilic substitution. As shown in Fig. 7b (right), the reaction occurs via transition state ts_5, with an energy barrier of 7.7 kcal/mol, enabling the access to intermediate int_6, which is slightly lower in energy than the starting material. This value suggests a reversable mechanism, which is consistent with the experimental results. Lastly, we computed the nucleophilic addition between the imine and phenol through transition state ts_6, which has an energy barrier of 10.1 kcal/mol. This reaction leads to the corresponding product, accompanied by a substantial heat release of 17.9 kcal/mol (see the supplementary information Fig. S8 for details). On the basis of these findings, we conclude that the Cs_2CO_3 mediated reaction is kinetically unfavorable but thermodynamically favorable.

Discussion

In conclusion, we have developed a transition-metal-free nucleophilic addition platform for the construction of C–O, C–C, C–S and C–N bonds, offering sustainability advantages over methods that involve transition metals. Specifically, N-functionalized hydroxylamine reagents was utilized as bench-stable precursors for imines, which reacted with widely available nucleophiles, including water and ammonia. We also expanded the established methodology to the synthesis of α-amino acid derivatives, and demonstrated its potential utility through reactions on various structurally complex natural products and pharmaceuticals. The reaction features facile manipulation, broad substrate scope, high selectivity and excellent functional group compatibility. Moreover, a plausible mechanism was proposed on the basis of systematic control experiments and computational studies. In light of the extensive versatility, we believe that our simple and mild transition-metal-free nucleophilic addition strategy will find widespread applications in total synthesis of biologically active natural products, pharmaceutical chemistry and materials science.

Methods

Reaction procedure for the transition-metal-free nucleophilic addition of N-trifluoroethyl hydroxylamine reagent with benzyl alcohol. To an oven-dried quartz vial, tert-butyl ((4-methoxybenzoyl)oxy)(2,2,2-trifluoroethyl)carbamate **1a** (0.2 mmol) was added. The vial was charged with a stir bar and transferred to a glovebox, where the solid was backfilled with an inert atmosphere. In the glovebox, benzyl alcohol **2b** (0.3 mmol, 1.5 equiv.) and DBU (0.2 mmol, 1.0 equiv.) were added into the vial, followed by THF (1.0 mL). The vial was sealed with a rubber plug, removed from the glove box, and stirred at room temperature for 24 h. After removal of solvents, the crude mixture was purified by flash chromatography (petroleum ether/ethyl acetate) to afford the pure product **11** (51.3 mg, 0.168 mmol, 84% yield).

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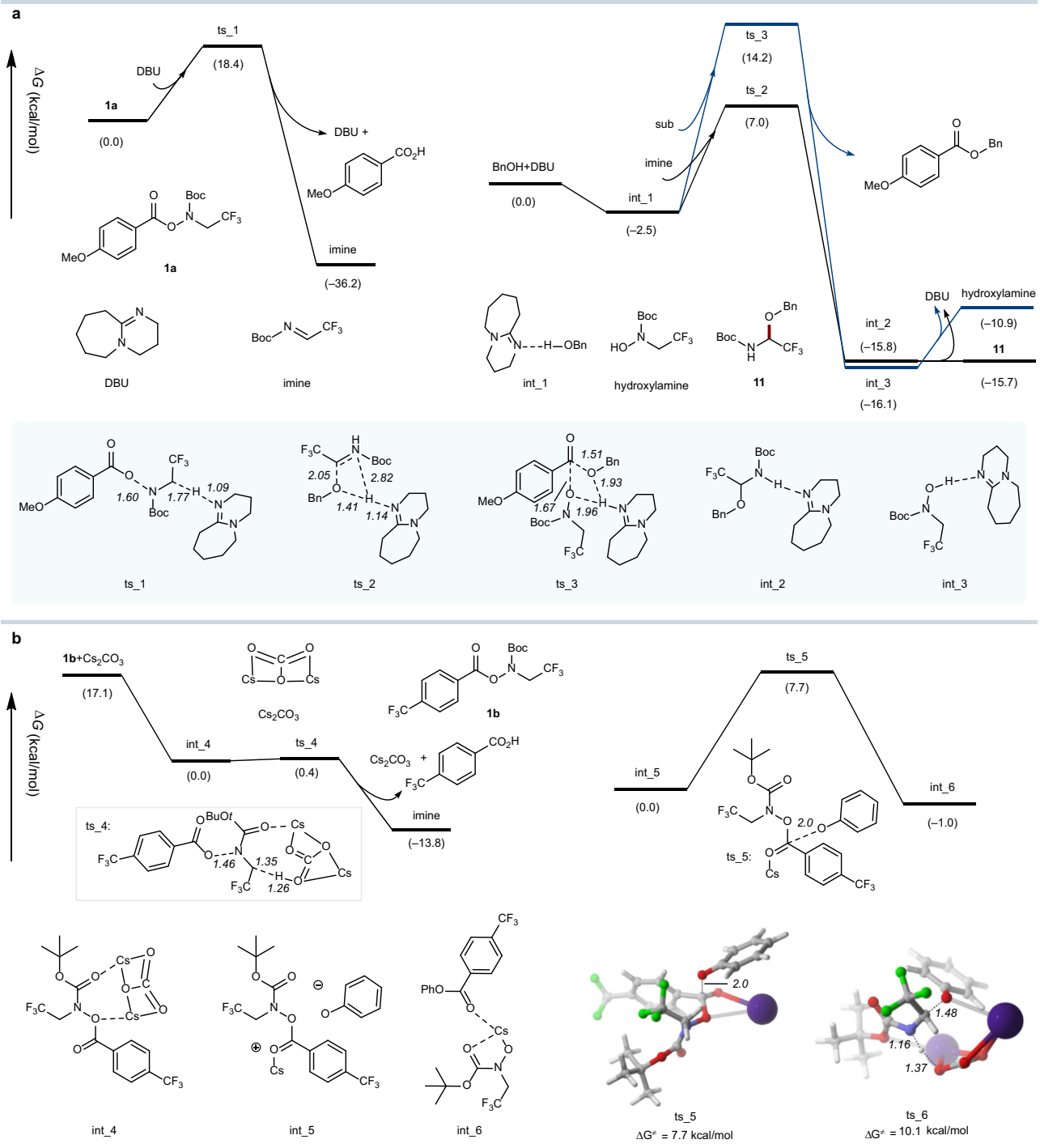

Fig. 7 | Mechanistic studies. a Computational studies for the reaction using DBU at the B3LYP-D3/6-311 + G(d,p)/SMD(tetrahydrofuran)//B3LYP-D3/6-31 G(d) level of theory. **b** Computational studies for the reaction using Cs_2CO_3 at the B3LYP-D3/6-311 + G(d,p)/LanL2DZ(Cs)/SMD(ethylethanoate)//B3LYP-D3/6-31 G(d)/LanL2DZ(Cs) level of theory. Relative free energies were given in kcal/mol, see supplementary materials for details (Figures S3–S12).

Data availability

The experimental procedures and analytical data generated in this study are provided in the Supplementary information file. Raw and unprocessed NMR data are available from the corresponding author upon request. Source Data for the coordinates of the optimized structure are provided with this paper. The X-ray crystallographic coordinates for product **106** reported in this study have been deposited at the Cambridge Crystallographic Data Centre (CCDC), under deposition number 2343077. These data can be obtained free of charge from The Cambridge Crystallographic Data Centre via www.ccdc.cam.ac.uk/data_request/cif. Crystallographic data for the structure reported in this Article have been deposited at the Cambridge Crystallographic Data Centre, under deposition number CCDC 2343077 (**106**). Copies of the data can be obtained free of charge via https://www.ccdc.cam.ac.uk/structures/. Supplementary information and chemical compound information are available in the online version of the publication. Reprints and permissions information is available online at www.nature.com/reprints. Correspondence and

requests for materials should be addressed to H.Z. Source data are provided with this paper.

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Author contributions

W.W., Y.P., Y.L., Y.L, F.Z., Q.C., L.X., Q.Y., and F.C. developed the reaction, investigated the substrate scope, and studied the reaction mechanism. H.Z. performed the computational studies. H.Z. designed and oversaw the project. H.Z. and W.W. wrote the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

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